

Underlay Physical Port / IP Mapping

LEAF1 E1 10.0.0.0/31 <-> SPINE1 E1 10.0.0.1/31
LEAF1 E2 10.0.0.12/31 <-> SPINE2 E1 10.0.0.13/31
LEAF2 E1 10.0.0.2/31 <-> SPINE1 E2 10.0.0.3/31
LEAF2 E2 10.0.0.14/31 <-> SPINE2 E2 10.0.0.15/31
LEAF3 E1 10.0.0.4/31 <-> SPINE1 E3 10.0.0.5/31
LEAF3 E2 10.0.0.16/31 <-> SPINE2 E3 10.0.0.17/31
LEAF4 E1 10.0.0.6/31 <-> SPINE1 E4 10.0.0.7/31
LEAF4 E2 10.0.0.18/31 <-> SPINE2 E4 10.0.0.19/31
BL1 E1 10.0.0.8/31 <-> SPINE1 E5 10.0.0.9/31
BL1 E2 10.0.0.20/31 <-> SPINE2 E5 10.0.0.21/31
BL2 E1 10.0.0.10/31 <-> SPINE1 E6 10.0.0.11/31
BL2 E2 10.0.0.22/31 <-> SPINE2 E6 10.0.0.23/31
RR1 E1 10.0.100.1/31 <-> SPINE1 E10 10.0.100.0/31
RR1 E2 10.0.100.3/31 <-> SPINE2 E10 10.0.100.2/31
RR2 E1 10.0.100.5/31 <-> SPINE1 E11 10.0.100.4/31
RR2 E2 10.0.100.7/31 <-> SPINE2 E11 10.0.100.6/31

Fabric Roles / Loopbacks / VTEP

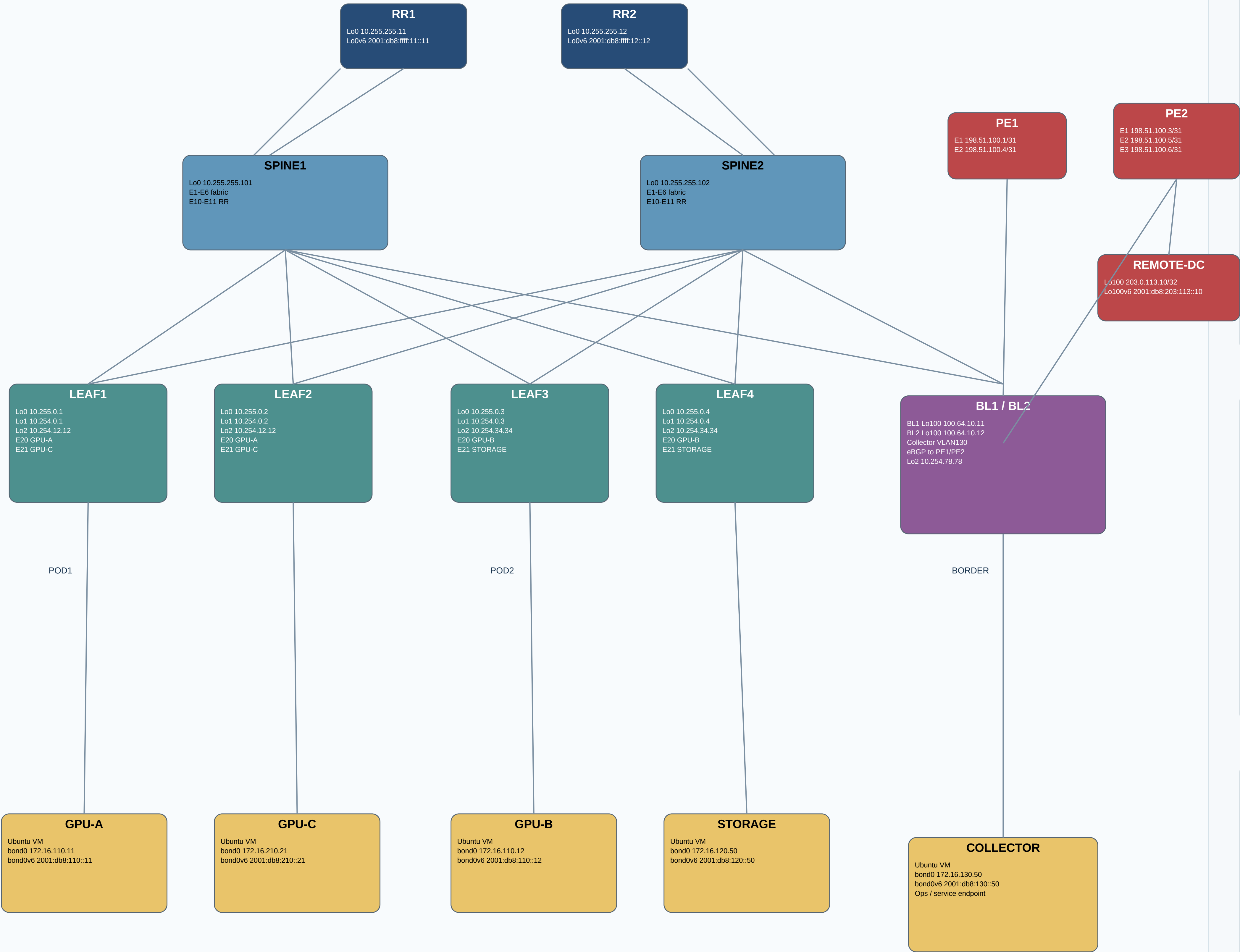
RR1 Lo0 10.255.255.11 | RR2 Lo0 10.255.255.12
SPINE1 Lo0 10.255.255.101 | SPINE2 Lo0 10.255.255.102
LEAF1 Lo0 10.255.0.1 | Lo1 10.254.0.1 | Lo2 10.254.12.12
LEAF2 Lo0 10.255.0.2 | Lo1 10.254.0.2 | Lo2 10.254.12.12
LEAF3 Lo0 10.255.0.3 | Lo1 10.254.0.3 | Lo2 10.254.34.34
LEAF4 Lo0 10.255.0.4 | Lo1 10.254.0.4 | Lo2 10.254.34.34
BL1 Lo1 10.254.0.11 | BL2 Lo1 10.254.0.12
BL1/BL2 shared border VTEP Lo2 10.254.78.78
RR1/RR2 act as EVPN route reflectors
Spines carry underlay transit only
Leaf pairs are MLAG + anycast gateway nodes
BL1/BL2 provide services access and eBGP edge

Host / Border Physical Ports

GPU-A eth0 -> LEAF1 E20 | eth1 -> LEAF2 E20 | Po10 | VLAN110
GPU-C eth0 -> LEAF1 E21 | eth1 -> LEAF2 E21 | Po11 | VLAN210
GPU-B eth0 -> LEAF3 E20 | eth1 -> LEAF4 E20 | Po20 | VLAN110
STORAGE eth0 -> LEAF3 E21 | eth1 -> LEAF4 E21 | Po21 | VLAN120
COLLECTOR eth0 -> BL1 E20 | eth1 -> BL2 E20 | Po30 | VLAN130
LEAF1 E10-E11 <-> LEAF2 E10-E11 | Po100 peer-link
LEAF3 E10-E11 <-> LEAF4 E10-E11 | Po100 peer-link
BL1 E10-E11 <-> BL2 E10-E11 | Po100 peer-link
BL1 E30 198.51.100.0/31 <-> PE1 E1 198.51.100.1/31
BL2 E30 198.51.100.2/31 <-> PE2 E1 198.51.100.3/31
PE1 E2 198.51.100.4/31 <-> PE2 E2 198.51.100.5/31
PE2 E3 198.51.100.6/31 <-> REMOTE E1 198.51.100.7/31

Physical Topology

central drawing intentionally simplified; exact port/IP assignments are in the side tables



Dual-Stack VLAN / VNI / Gateway Mapping

VLAN110 | VNI10110 | VRF AI_FABRIC
IPv4 172.16.110.0/24 | GW 172.16.110.1
IPv6 2001:db8:110::/64 | GW 2001:db8:110::1
VLAN120 | VNI10120 | VRF AI_FABRIC
IPv4 172.16.120.0/24 | GW 172.16.120.1
IPv6 2001:db8:120::/64 | GW 2001:db8:120::1
VLAN130 | VNI10130 | VRF AI_SERVICES
IPv4 172.16.130.0/24 | GW 172.16.130.1
IPv6 2001:db8:130::/64 | GW 2001:db8:130::1
VLAN210 | VNI10210 | VRF TENANT_B
IPv4 172.16.210.0/24 | GW 172.16.210.1
IPv6 2001:db8:210::/64 | GW 2001:db8:210::1

Endpoint Addressing

GPU-A Ubuntu VM
bond0 172.16.110.11 | bond0v6 2001:db8:110::11
GPU-B Ubuntu VM
bond0 172.16.110.12 | bond0v6 2001:db8:110::12
GPU-C Ubuntu VM
bond0 172.16.210.21 | bond0v6 2001:db8:210::21
STORAGE Ubuntu VM
bond0 172.16.120.50 | bond0v6 2001:db8:120::50
COLLECTOR Ubuntu VM
bond0 172.16.130.50 | bond0v6 2001:db8:130::50
REMOTE-DC
Lo100 203.0.113.10/32 | 2001:db8:203:113::10/128

Topology Notes

GPU nodes are Ubuntu VMs, not switches.
RoCE study focus uses Ethernet fabric behavior in EVE-NG.
InfiniBand concepts remain workbook-guided, not fully emulated.
Anycast gateways live on MLAG leaf pairs and border pair.
Spines do not host tenant SVIs.
Use these tables as the source of truth for physical mapping.
Topology center is intentionally uncluttered for readability.
Use the workbook for config, validation, and troubleshooting tasks.